ABOUT / CURRICULUM VITAE

Yuning Lei, Ph.D.

Research Associate in Microrobotics and Control Engineering at the University of Oldenburg, Germany.

I have been a Research Associate at the University of Oldenburg since April 2023. My research focuses on liquid-metal nanocharacterization, microassembly in SEM-FIB and nanorobotic structures.

I received my Ph.D. in Engineering Science from Université Bourgogne Franche-Comté in 2023, following doctoral research at FEMTO-ST Institute. I previously studied at the University of Orléans.

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Academic record as of 20 October 2026.

Research & experience

April 2023 – present

Research Associate

University of Oldenburg · Oldenburg, Germany

- Microrobotics and Control Engineering, Department of Computing Science.
- Research project PI for LiCoVac (Liquid Metal Contact Angle Measurement in Vacuum).
- Planning and management of international collaborative research and staff exchanges (EU-HORIZON-MSCA, DAAD-PPP).

October 2019 – March 2023

Ph.D. Fellow

FEMTO-ST Institute · Besançon, France

- Silica origami fabrication, robotic microassembly, characterization and actuation at the tip of an optical fiber in SEM.
- Advisors: Cédric Clévy, Jean-Yves Rauch and Philippe Lutz.
- Contributions to the COLAMIR and DyNaBot collaborative research projects.

March – September 2019

Laboratory Trainee

PRISME Laboratory · Orléans, France

- Data assimilation of parallel turbulent flows from experimental local measurements, using linearized Navier–Stokes equations and Kalman-filter-based feedback control.

April 2016 – September 2017

Laboratory Apprentice Technician

WILO Intec · Aubigny-sur-Nère, France

- Hydraulic testing of centrifugal pumps, measurement uncertainty analysis and development of an uncertainty-analysis tool.

Education

October 2019 – March 2023

Ph.D. in Engineering Science

Université Bourgogne Franche-Comté · Besançon, France

- Dissertation: Nanorobotic Origami Unfolded at the Tip of Optical Fiber.
- Defended on 24 March 2023 in Besançon.

September 2017 – September 2019

M.Sc. in Mechatronics, Autonomous, Robotics and Signal

University of Orléans · Orléans, France

- Robotic vision, visual servoing, mobile-robot control, complex-system modeling and feedback control.

September 2016 – September 2017

B.Sc. in Expertise Environmental Measurements

IUT de Bourges, University of Orléans · Bourges, France

- Environmental monitoring, physicochemical analysis, chromatography and spectrometry.

September 2014 – September 2016

University Technology Degree (DUT) in Physical Measurements

IUT de Bourges, University of Orléans · Bourges, France

September 2013 – June 2014

First-year studies in Computer Science

IUT de Lannion, University of Rennes · Lannion, France

- Completed the first year before switching to Physical Measurements.

Lectures & tutorials

Spring 2024–2026

Force Sensing (Microrobotics I)

AMiR, University of Oldenburg

Lecture · 1.5 h × 1 session

Spring 2024–2026

Scanning Probe Microscope (Microrobotics II)

AMiR, University of Oldenburg

Lecture · 1.5 h × 1 session

Spring 2024–2026

AFM in SEM (Microrobotics II)

AMiR, University of Oldenburg

Lecture · 1.5 h × 1 session

Spring 2023

Automated Nanohandling (Microrobotics II)

AMiR, University of Oldenburg

Lectures · 1.5 h × 2 sessions

Fall 2021 & 2022

Digital Control

Master MIR, UFR-ST

Tutorials · 4 h × 5 sessions

Fall 2020 & 2021

Automation and Control

ISIFC, UFC

Tutorials · 4 h × 8 sessions

Fall 2020 & 2021

Servo-control with MATLAB and Simulink

Master SAPIAA, UFR-ST

Tutorials · 1.5 h × 4 sessions

Technical skills

SEM / FIB	Zeiss Auriga 60, Thermo Fisher APREO S, FEI Helios NanoLab, FEI Quanta 600 and Tescan Lyra2.
Microfabrication	Lithography, physical vapor deposition (PVD), wet etching and plasma etching (RIE/DRIE).
Characterization	Nanoindentation, interferometry and profilometry.
Robotics & computing	MATLAB/Simulink, LabVIEW, ROS, V-REP; C/C++, Java, Python, MATLAB and Visual Basic.
Design & simulation	SolidWorks, Creo, AutoCAD/Inventor, Blender, Inkscape, Alphacam; COMSOL and ANSYS.

Service & memberships

Since November 2020 — **MARSS Community**
Publicity Chair and Program Committee member

Since February 2021 — **IEEE Robotics and Automation Society**
Member

Since November 2020 — **SAGIP**
Member

Since September 2017 — **GDR Robotique**
Member

Since November 2020 — **GDR MACS**
Member

Honors & awards

September 2022 — **“Micron d’Or” Award**
Salon Micronora 2022, Association Micronora

March 2022 — **3rd Award — Photo Contest**
CAZAC Day 2022, Carl Zeiss Application Community

July 2020 — **“Breakthrough in Science” Award**
Ph.D. Student Poster Competition, FEMTO-ST Institute

CNU qualification

March 2026 — **CNU qualification, Section 61**
Qualification for Associate Professor positions in French universities.

Languages

Chinese	Native
French	Advanced
English	B1; TOEIC 720/990 (December 2017)
German	Elementary

Publications

Reproducible fabrication of liquid metal droplets via electromigration and FIB sputtering in SEM

Yuning Lei; James L. Mead; Waldemar Klauser; Sergej Fatikow
Journal of Micro and Bio Robotics, 21, article 14, 2025.
https://doi.org/10.1007/s12213-025-00192-5

Reproducible Fabrication of Liquid Metal Droplet with FIB Sputtering for Contact Angle Measurement in SEM

Yuning Lei; Sergej Fatikow

2024 International Conference on Manipulation, Automation and Robotics at Small Scales (MARSS), pp. 1–6, 2024.

https://doi.org/10.1109/MARSS61851.2024.10612731

High resolution surface contact temperature measurements by means of micro-thermocouples in vacuum conditions

L. Thiery; J.Y. Rauch; **Y. Lei**

International Journal of Thermal Sciences, 195, article 108663, 2024.

https://doi.org/10.1016/j.ijthermalsci.2023.108663

Large-Workspace Polyarticulated Micro-Structures Based-On Folded Silica for Tethered Nanorobotics

Yuning Lei; Cédric Clévy; Jean-Yves Rauch; Philippe Lutz

IEEE Robotics and Automation Letters, 7(1), pp. 88–95, 2022.

https://doi.org/10.1109/LRA.2021.3118470